

Life Processes

Science • Chapter 5 • Chapter Summary & Study Guide for Daksh

What makes a living organism 'alive' — even when it appears motionless? The answer lies in a set of basic **life processes** that go on inside every living thing: **nutrition, respiration, transport (circulation), and excretion**. This chapter explores how plants and animals carry out each of these — sometimes very differently — to maintain themselves.

Nutrition — getting energy and building blocks

Two broad modes:

- **Autotrophic** nutrition (Greek: 'self-feeding') — plants and some bacteria make their own food by **photosynthesis** using sunlight, CO_2 and water in chlorophyll-containing chloroplasts. Equation: $6\text{CO}_2 + 6\text{H}_2\text{O} \xrightarrow{\text{sunlight, chlorophyll}} \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$.
- **Heterotrophic** nutrition — animals depend on others. Sub-types: **holozoic** (Amoeba, humans — ingest, digest, absorb), **saprophytic** (fungi, bread mould — feed on dead matter externally), **parasitic** (Cuscuta, tapeworm — live on host).

Human digestive system: mouth → oesophagus → stomach → small intestine → large intestine → anus, with associated glands — salivary glands, liver and pancreas. Food is broken down by **enzymes**:

Mouth : Salivary amylase (ptyalin) → starch → maltose
Stomach : HCl + pepsin → proteins → peptones (acidic medium)
Sm. intestine: Bile (liver) emulsifies fat;
Pancreatic juice (trypsin, lipase, amylase);
Intestinal juice → simple sugars, amino acids, fatty acids.
Absorption: through villi (large surface area, blood + lymph) into circulation.

Respiration — releasing energy from food

All cells need energy in the form of **ATP**, released by oxidising glucose. Two routes:

- **Aerobic** (with O_2): Glucose → CO_2 + H_2O + much energy. Happens in mitochondria.
- **Anaerobic** (without O_2): Glucose → ethanol + CO_2 (in yeast, fermentation), or → lactic acid (in our muscles during heavy exercise — causing cramps), with less energy.

Human respiratory system: nostrils → pharynx → larynx → trachea → bronchi → bronchioles → alveoli (in lungs). Alveoli are millions of thin-walled balloon-like sacs that maximise surface area for gas exchange. Oxygen diffuses into blood capillaries, binds to **haemoglobin** in RBCs as oxyhaemoglobin; CO_2 diffuses out and is exhaled. Diaphragm + intercostal muscles drive breathing. Plants use stomata for gas exchange.

Transportation — moving substances around

In humans a **double circulation** (pulmonary + systemic) keeps oxygenated and deoxygenated blood separate. The heart has 4 chambers — left and right atria (receive) and ventricles (pump). Blood vessels: **arteries** (thick walls, carry away from heart, mostly oxygenated except pulmonary), **veins** (thin walls, valves, carry to heart), and **capillaries** (one-cell thick, exchange site). The **lymphatic system** drains tissue fluid back to circulation.

In plants there are two transport tissues: **xylem** carries water and minerals upward from roots (driven by transpiration pull) and **phloem** transports food (sucrose) from leaves to all parts (translocation, powered by ATP).

Excretion — removing wastes

Metabolic wastes (urea, uric acid, excess salts) must be removed to keep the internal environment stable (homeostasis). In humans, the **kidneys** contain ~1 million **nephrons** each — the structural and functional unit. Filtration occurs at the **glomerulus** inside the Bowman's capsule under pressure; useful things (glucose, water, salts) are reabsorbed in the tubule; the rest becomes urine, collected via the ureters into the bladder. **Dialysis**

mimics this when kidneys fail.

Plants get rid of waste by storage (resin, gums in old xylem), by losing leaves (containing wastes), by diffusing gaseous wastes (O_2 , CO_2) through stomata, and by releasing some wastes into the soil through roots.

★ MUST-REMEMBER POINTS

- Photosynthesis: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (in chloroplast, with chlorophyll & sunlight).
- Aerobic respiration releases more energy than anaerobic.
- Lactic acid build-up in muscles during heavy exercise causes cramps.
- Heart has 4 chambers; double circulation = pulmonary + systemic.
- Xylem carries water/minerals (one-way, up); Phloem carries food (both directions).
- Nephron is the functional unit of kidney; filtration at glomerulus, reabsorption in tubule.
- Stomata in leaves = sites of gaseous exchange and transpiration.

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